

Draw the Indian Flag With Python Turtle!

Learn to paint with code lines and make a beautiful glowing flag appear.



Watch the beautiful bands and the spinning chakra come alive!

Hello Little Coding Artist! Welcome to your magical canvas. Today, we are going to learn how to guide a friendly digital turtle across your computer screen to sketch and fill in our proud national flag!

The Magic Flag Screen Adventure

Imagine launching your program and seeing three gorgeous, neat color bands paint themselves one by one across a clean canvas. Then, a beautiful blue **Ashoka Chakra** spins its lines perfectly right in the center!



What We Will Code Today:

- ★ **Meet the Turtle:** Get to know your virtual drawing pet.
- ★ **The Three Stripes:** Draw equal boxes for Saffron, White, and Green.
- ★ **The Loop Shortcut:** Use a trick to draw 24 lines without getting tired.
- ★ **Color Fills:** Learn how to splash liquid color into your shapes.

You don't need real paint brushes or messy liquids today. Your keyboard buttons are going to be your magic artistic tools!

Meet Your Coding Friend: The Turtle!

In Python, we have an awesome friend who loves to draw for us. This friend is a little digital pet called the **Turtle**!

Walking on a Game Board:

Imagine your turtle walks step by step across a giant wooden game board. Wherever his tiny feet walk, he leaves a colored ink path behind him! You are the boss of the turtle. He only moves when you type a command spell!

Your Four Movement Spells:

- ★ `forward(50)` → Tells the turtle to walk straight ahead by 50 little steps.
- ★ `backward(50)` → Tells the turtle to back up without turning around.
- ★ `right(90)` → Tells the turtle to pivot like turning a corner in a game of Kabaddi!
- ★ `left(90)` → Tells the turtle to spin around to his left side.

The Flying Pen Tricks: PenUp & PenDown

If the turtle leaves a mark everywhere he goes, how do we move him to a new corner without creating an ugly, messy streak across our beautiful flag?



Lift Your Sketch Pen!

Imagine you are holding a bright orange sketch pen over a white drawing sheet. If you want to draw a circle on the right side, you lift your hand up, move through the air, and press it back down on the paper. Python does this using two spells:

```
penup()  
forward(100)  
pendown()
```

Using `penup()` lifts the turtle's ink tail off the canvas. He can fly through the air completely invisible! When he lands, calling `pendown()` drops his tail back down so he can draw again.

Painting the Three Equal Bands

The Indian Flag has three beautiful horizontal bands. To make it look perfect, each band must have the exact same width and thickness!



Our Three Secret Color Names:

- **Top Band:** Saffron Orange (We use a special code: "#FF9933")
- **Middle Band:** Clean White
- **Bottom Band:** Deep Festive Green

How to Fill a Shape with Color:

To fill a stripe with color, you must warn the turtle before you start drawing the outline box, and notify him when you are finished:

```
color("#FF9933")
begin_fill()
# (Draw your rectangle box here)
end_fill()
```

The Ashoka Chakra Ferris Wheel!

Right in the center of our white stripe sits the navy blue **Ashoka Chakra**. It has 24 beautiful lines coming out from the center point, just like the spokes of a giant bicycle wheel or a carnival Ferris Wheel!

The Lazy Coder Problem

Imagine typing out the commands to draw a line, turn, back up, and move again 24 separate times. Your fingers would get super tired, and your script would look way too long!

The Magic 'For Loop' Shortcut

Instead of copying and pasting the lines, we can tell Python to repeat a block of code automatically using a tool called a **For Loop**. It acts like a fast repeating machine that handles the repetitive work in a split second!

How to Spin the Chakra Spokes

Let's look at the actual code block that draws all 24 spokes smoothly. It uses our loop machine and a little bit of circle math geometry:

```
color("navy")
for i in range(24):
    forward(30)
    backward(30)
    right(15)
```

What Each Spacing Step Does:

- **forward(30):** The turtle walks out from the center to draw a spoke line.
- **backward(30):** He steps backward along the same line to return to the center.
- **right(15):** He turns slightly to his right by 15 degrees. Why 15? Because a full circle has 360 degrees, and 360 divided by 24 spokes equals 15!

Bug Hunting: The Invisible Turtle Mystery!

Sometimes, you run your code script and look at the screen, but nothing shows up! It's completely blank. Oh no! Let's find out where the hidden bugs are sleeping.

Check These 3 Classic Bug Traps:

1. **The Floating Pen:** Did you call `penup()` to fly, but forgot to call `pendown()` when you landed? If the pen is up, the turtle draws with invisible ink!
2. **The Half-Eaten Fill:** Did you call `begin_fill()` but forgot to type `end_fill()` at the end? The turtle will hold the paint bottle and never pour it out!
3. **Missing Brackets:** Typing `forward 50` without the round brackets `()` makes Python confused. Always use brackets!

Putting the Whole Flag Together!

Let us review the exact plan your program will execute from start to finish. It's exactly like completing a beautiful connect-the-dots sticker book:

The Complete Flag Recipe Steps:

- ★ **Step 1:** Start at the top left, select Saffron, fill in rectangle #1.
- ★ **Step 2:** Move the pen down without drawing, select White, fill in rectangle #2.
- ★ **Step 3:** Move down again, select Green, fill in rectangle #3.
- ★ **Step 4:** Go to the absolute center of the flag, select Navy Blue.
- ★ **Step 5:** Run your 24-step loop wheel to draw the gorgeous Ashoka Chakra spokes!

Congratulations, Master Flag Designer!

Hooray! You have learned how to combine math, logic colors, and loops to paint our proud national symbol using code lines.

 **MISSION ACCOMPLISHED!** 

You can control turtle paths, toggle pens, and spin loop wheels.

 **Official Python Turtle Artist Badge!** 

**Fire up your code screen, type out your turtle commands,
and show your beautiful artwork to your friends and family!**